

```
In [ ]: """
Data Cleaning Concepts to use to resolve this problem.  Use the AB_NYC dataset (CSV file) attached to Week4 folder.

Load the Airbnb dataset into a Pandas DataFrame.
Randomly sample 1,000 rows from the dataset.

Identify missing values in the dataset.
Apply an appropriate cleaning strategy:
Detect outliers in price, minimum_nights
Decide whether to remove, keep outliers

Identify irrelevant or redundant columns.
Remove at least three attributes and justify why.
Show me / Reduce data to only show attributes useful for predicting listing price

Most frequently (ie. 80% of the time) used Python statements for data cleaning purposes are:

df.info()

df.isna().sum()

dropna() / fillna()

SQL filtering

drop_duplicates()

There are many others.  Experiment around with various Python statements that are used for data cleaning.
"""
```

```
In [3]: import pandas as pd

# Load dataset
```

```
df = pd.read_csv(r"C:\Users\kesha\Downloads\AB_NYC_2019.csv")

# Display basic information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48895 entries, 0 to 48894
Data columns (total 16 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     48895 non-null  int64
1   name                                  48879 non-null  object
2   host_id                               48895 non-null  int64
3   host_name                             48874 non-null  object
4   neighbourhood_group                   48895 non-null  object
5   neighbourhood                         48895 non-null  object
6   latitude                             48895 non-null  float64
7   longitude                             48895 non-null  float64
8   room_type                             48895 non-null  object
9   price                                 48895 non-null  int64
10  minimum_nights                        48895 non-null  int64
11  number_of_reviews                     48895 non-null  int64
12  last_review                           38843 non-null  object
13  reviews_per_month                     38843 non-null  float64
14  calculated_host_listings_count        48895 non-null  int64
15  availability_365                       48895 non-null  int64
dtypes: float64(3), int64(7), object(6)
memory usage: 6.0+ MB
```

```
In [4]: sample_df = df.sample(n=1000, random_state=42)

print(sample_df.shape)
```

```
(1000, 16)
```

```
In [5]: sample_df.isna().sum()
```

```
Out[5]: id          0
        name        0
        host_id     0
        host_name    0
        neighbourhood_group  0
        neighbourhood  0
        latitude     0
        longitude    0
        room_type    0
        price        0
        minimum_nights  0
        number_of_reviews  0
        last_review   214
        reviews_per_month  214
        calculated_host_listings_count  0
        availability_365  0
        dtype: int64
```

```
In [9]: sample_df["reviews_per_month"] = sample_df["reviews_per_month"].fillna(0)
        sample_df["host_name"] = sample_df["host_name"].fillna("Unknown")
        sample_df["name"] = sample_df["name"].fillna("No Name")
        print(sample_df["reviews_per_month"])
        print(sample_df["host_name"])
        print(sample_df["name"])
```

```

879      0.71
44383    0.00
15394    0.43
43230    1.88
16332    0.80
...
18421    2.52
9146     0.02
36356    0.00
39865    4.21
27627    0.20
Name: reviews_per_month, Length: 1000, dtype: float64
879      Jane
44383    Giancarlo
15394    Alejandro
43230    Sonder (NYC)
16332    Gingie
...
18421    Caroline
9146     Keith
36356    Kathy
39865    Tiba
27627    Ashley
Name: host_name, Length: 1000, dtype: object
879      Come and go as you please in BKLN!
44383    Spacious, sunny room in Queens/Brooklyn
15394    Private bedroom in high-ceiling 4BR apartment!
43230    Sonder | Stock Exchange | Stunning 3BR + Kitchen
16332    Spacious 2 Bedroom with Balcony
...
18421    Modern Apt - 10 min from Manhattan
9146     15 min walk to Prospect Park!
36356    Nice Couple in A Walk in Basement with Window
39865    Walk toTimes Square,Central Park,Museums, Broa...
27627    Bright East Village Apartment!
Name: name, Length: 1000, dtype: object

```

```

In [10]: sample_df["last_review"] = sample_df["last_review"].fillna("Not Reviewed")
         print(sample_df["last_review"])

```

```

879      2019-01-02
44383    Not Reviewed
15394      2017-04-28
43230      2019-06-02
16332      2019-06-03
...
18421      2019-06-13
9146      2015-09-17
36356    Not Reviewed
39865      2019-06-23
27627      2018-08-15
Name: last_review, Length: 1000, dtype: object

```

```

In [12]: sample_df.drop_duplicates(inplace=True)
         sample_df.duplicated().sum()

```

```

Out[12]: np.int64(0)

```

```

In [13]: Q1 = sample_df["minimum_nights"].quantile(0.25)
         Q3 = sample_df["minimum_nights"].quantile(0.75)

         IQR = Q3 - Q1

         lower = Q1 - 1.5 * IQR
         upper = Q3 + 1.5 * IQR

         night_outliers = sample_df[
             (sample_df["minimum_nights"] < lower) |
             (sample_df["minimum_nights"] > upper)
         ]

         print(len(night_outliers))

```

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```

In [14]: Q1 = sample_df["price"].quantile(0.25)
         Q3 = sample_df["price"].quantile(0.75)

         IQR = Q3 - Q1

         lower = Q1 - 1.5 * IQR
         upper = Q3 + 1.5 * IQR

```

```
price_outliers = sample_df[
    (sample_df["price"] < lower) |
    (sample_df["price"] > upper)
]

print(len(price_outliers))
```

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```
In [15]: sample_df.drop("id", axis=1, inplace=True)
sample_df.drop("host_id", axis=1, inplace=True)
sample_df.drop("host_name", axis=1, inplace=True)
```

```
In [16]: print(sample_df.drop)
```

```
<bound method DataFrame.drop of
879          Come and go as you please in BKLN!          Brooklyn
44383          Spacious, sunny room in Queens/Brooklyn      Queens
15394    Private bedroom in high-ceiling 4BR apartment!    Manhattan
43230    Sonder | Stock Exchange | Stunning 3BR + Kitchen  Manhattan
16332          Spacious 2 Bedroom with Balcony            Manhattan
...
18421          Modern Apt - 10 min from Manhattan          Queens
9146          15 min walk to Prospect Park!                Brooklyn
36356    Nice Couple in A Walk in Basement with Window    Brooklyn
39865    Walk toTimes Square,Central Park,Museums, Broa...  Manhattan
27627          Bright East Village Apartment!              Manhattan
```

```
          neighbourhood latitude longitude room_type price \
879          Kensington 40.64354 -73.97777 Entire home/apt 89
44383          Ridgewood 40.70666 -73.90779 Private room 30
15394          Hell's Kitchen 40.76116 -73.99016 Private room 120
43230          Financial District 40.70763 -74.01050 Entire home/apt 470
16332          East Harlem 40.79658 -73.93287 Entire home/apt 199
...
18421          East Elmhurst 40.75720 -73.89569 Entire home/apt 100
9146    Prospect-Lefferts Gardens 40.65887 -73.94882 Private room 30
36356          Borough Park 40.63416 -74.00722 Private room 50
39865          Hell's Kitchen 40.76356 -73.98956 Entire home/apt 230
27627          East Village 40.72411 -73.98872 Entire home/apt 99
```

```
          minimum_nights number_of_reviews last_review reviews_per_month \
879          3          62    2019-01-02          0.71
44383          21          0    Not Reviewed          0.00
15394          2          17    2017-04-28          0.43
43230          2          5    2019-06-02          1.88
16332          2          30    2019-06-03          0.80
...
18421          3          87    2019-06-13          2.52
9146          6          1    2015-09-17          0.02
36356          2          0    Not Reviewed          0.00
39865          1          17    2019-06-23          4.21
27627          5          4    2018-08-15          0.20
```

```
          calculated_host_listings_count availability_365
879          1          189
44383          1          73
```

15394	1	0
43230	327	272
16332	1	30
...	...	...
18421	1	322
9146	1	0
36356	10	365
39865	5	282
27627	1	0

[1000 rows x 13 columns]>

```
In [19]: price_df = sample_df[
    [
        "neighbourhood_group",
        "neighbourhood",
        "latitude",
        "longitude",
        "room_type",
        "minimum_nights",
        "number_of_reviews",
        "reviews_per_month",
        "calculated_host_listings_count",
        "availability_365",
        "price"
    ]
]
price_df.head()
df.info()
```



```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48895 entries, 0 to 48894
Data columns (total 16 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   id                  48895 non-null  int64
 1   name                48879 non-null  object
 2   host_id             48895 non-null  int64
 3   host_name           48874 non-null  object
 4   neighbourhood_group 48895 non-null  object
 5   neighbourhood        48895 non-null  object
 6   latitude            48895 non-null  float64
 7   longitude           48895 non-null  float64
 8   room_type           48895 non-null  object
 9   price               48895 non-null  int64
10  minimum_nights      48895 non-null  int64
11  number_of_reviews   48895 non-null  int64
12  last_review         38843 non-null  object
13  reviews_per_month   38843 non-null  float64
14  calculated_host_listings_count 48895 non-null  int64
15  availability_365     48895 non-null  int64
dtypes: float64(3), int64(7), object(6)
memory usage: 6.0+ MB

```

```
In [20]: df.isna().sum()
```

```
Out[20]: id          0
         name        16
         host_id     0
         host_name    21
         neighbourhood_group  0
         neighbourhood  0
         latitude     0
         longitude    0
         room_type    0
         price        0
         minimum_nights  0
         number_of_reviews  0
         last_review   10052
         reviews_per_month  10052
         calculated_host_listings_count  0
         availability_365  0
         dtype: int64
```

```
In [21]: df.dropna()
```

Out[21]:

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt
5	5099	Large Cozy 1 BR Apartment In Midtown East	7322	Chris	Manhattan	Murray Hill	40.74767	-73.97500	Entire home/apt
...	...	...	...	...	...	...	...	...	...
48782	36425863	Lovely Privet Bedroom with Privet Restroom	83554966	Rusaa	Manhattan	Upper East Side	40.78099	-73.95366	Private room
48790	36427429	No.2 with queen size bed	257683179	H Ai	Queens	Flushing	40.75104	-73.81459	Private room
48799	36438336	Seas The Moment	211644523	Ben	Staten Island	Great Kills	40.54179	-74.14275	Private room

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type
48805	36442252	1B-1B apartment near by Metro	273841667	Blaine	Bronx	Mott Haven	40.80787	-73.92400	Entire home/apt
48852	36455809	Cozy Private Room in Bushwick, Brooklyn	74162901	Christine	Brooklyn	Bushwick	40.69805	-73.92801	Private room

38821 rows × 16 columns

```
In [22]: df.fillna(0)
```

Out[22]:

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/ap
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/ap
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/ap
...	...	...	...	...	...	...	...	...	...
48890	36484665	Charming one bedroom - newly renovated rowhouse	8232441	Sabrina	Brooklyn	Bedford-Stuyvesant	40.67853	-73.94995	Private room
48891	36485057	Affordable room in Bushwick/East Williamsburg	6570630	Marisol	Brooklyn	Bushwick	40.70184	-73.93317	Private room
48892	36485431	Sunny Studio at Historical Neighborhood	23492952	Ilgar & Aysel	Manhattan	Harlem	40.81475	-73.94867	Entire home/ap
48893	36485609	43rd St. Time Square-cozy single bed	30985759	Taz	Manhattan	Hell's Kitchen	40.75751	-73.99112	Shared room

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type
48894	36487245	Trendy duplex in the very heart of Hell's Kitchen	68119814	Christophe	Manhattan	Hell's Kitchen	40.76404	-73.98933	Private room

48895 rows × 16 columns

```
In [23]: df.drop_duplicates()
```

Out[23]:

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/ap
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/ap
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/ap
...	...	...	...	...	...	...	...	...	...
48890	36484665	Charming one bedroom - newly renovated rowhouse	8232441	Sabrina	Brooklyn	Bedford-Stuyvesant	40.67853	-73.94995	Private room
48891	36485057	Affordable room in Bushwick/East Williamsburg	6570630	Marisol	Brooklyn	Bushwick	40.70184	-73.93317	Private room
48892	36485431	Sunny Studio at Historical Neighborhood	23492952	Ilgar & Aysel	Manhattan	Harlem	40.81475	-73.94867	Entire home/ap
48893	36485609	43rd St. Time Square-cozy single bed	30985759	Taz	Manhattan	Hell's Kitchen	40.75751	-73.99112	Shared room

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type
48894	36487245	Trendy duplex in the very heart of Hell's Kitchen	68119814	Christophe	Manhattan	Hell's Kitchen	40.76404	-73.98933	Private room

48895 rows × 16 columns

In []: